

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Comparative Analysis

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO4: *Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)*

Assessment Tool Weightage: 10%

Compute the required performance measures from the given data and compare the techniques based on the results. Evaluate and justify your conclusion using the calculated values.

1. Detection Performance Comparison

The two object detection models are given as: Viola-Jones (Model A) and YOLO (Model B).

Given Data :

Model	TP	FP	FN
A – Viola-Jones	80	20	30
B – YOLO	90	15	20

Formula

Precision

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall

$$\text{Recall} = \frac{TP}{TP + FN}$$

Model A – Viola-Jones

Precision:

$$80/20+80=80/100=0.80$$

Precision = 80%

Recall:

$$80/80+30 = 80/110 = 0.7273$$

Recall = 72.73%

Model B – YOLO

Precision:

$$90/90+15=90/105=0.8571$$

Precision = 85.71%

Recall:

$$90/90+20 = 90/110 = 0.8182$$

Recall = 81.82%

Comparison

YOLO has higher **precision (85.71%)** than Viola-Jones (80%). It also has higher **recall (81.82%)** than Viola-Jones (72.73%).

Conclusion

YOLO is preferable because it produces fewer false positives and false negatives while achieving better precision and recall. Therefore, YOLO provides better overall object detection performance.

5. Pattern Recognition Accuracy Comparison

The assessment gives the correct predictions and total samples for Traditional and Deep Learning systems.

Given Data :

System	Correct Predictions	Total Samples
Traditional	85	100
Deep Learning	92	100

Formula

Accuracy = (Correct Predictions / Total Samples) × 100

Traditional System:

$$\text{Accuracy}=(85/100)\times 100= 85\%$$

Deep Learning System:

Accuracy = $(92/100) \times 100 = 92\%$

Comparison:

Traditional system has an accuracy of 85%, while Deep Learning has an accuracy of 92%.

Conclusion:

Deep Learning performs better because it has a higher accuracy of 92% compared to 85% for the Traditional system. Therefore, Deep Learning is preferable for pattern recognition

7. F1-Score Comparison**Given Data:**

Model A:

Precision = 0.80

Recall = 0.60

Model B:

Precision = 0.75

Recall = 0.75

Formula:

F1-Score = $(2 \times \text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$

Model A:

$F1 = (2 \times 0.80 \times 0.60) / (0.80 + 0.60)$

$= 0.96 / 1.40$

$= 0.686$

$= \mathbf{68.57\%}$

Model B:

$F1 = (2 \times 0.75 \times 0.75) / (0.75 + 0.75)$

$= 1.125 / 1.50$

$= 0.75$

$= \mathbf{75\%}$

Comparison:

Model A has an F1-score of 68.57%, while Model B has an F1-score of 75%.

Conclusion:

Model B is better because it has a higher F1-score of 75%. It provides a better balance between precision and recall. Therefore, Model B is preferable.

10. Error Rate Comparison

Given Data:

MethodA:

Errors=20

Total = 100

MethodB:

Errors=10

Total = 100

Formula:

Error Rate = (Errors / Total) × 100

Method A:

ErrorRate=(20/100)×100= **20%**

Method B:

ErrorRate=(10/100)×100= **10%**

Comparison:

Method A has an error rate of 20%, while Method B has an error rate of 10%.

Conclusion:

Method B is better because it has a lower error rate of 10%. Lower error rate indicates better reliability and performance. Therefore, Method B is preferable.

12. IoU Comparison

Given Data:

Object1:

ModelA=0.60

Model B = 0.80

Object2:

ModelA=0.70

Model B = 0.85

Object3:

ModelA=0.65

Model B = 0.90

Formula:

Average IoU = Sum of IoU values / Number of Objects

Model A:

Average IoU = (0.60 + 0.70 + 0.65) / 3

$$= 1.95 / 3$$
$$= \mathbf{0.65}$$

Model B:

$$\text{Average IoU} = (0.80 + 0.85 + 0.90) / 3$$
$$= 2.55 / 3$$
$$= \mathbf{0.85}$$

Comparison:

Model A has an average IoU of 0.65, while Model B has an average IoU of 0.85.

Conclusion:

Model B is better because it has a higher average IoU of 0.85. Higher IoU indicates better overlap between the predicted and actual object regions. Therefore, Model B provides better detection quality

Code of Conduct:

I V G ABBIRAMY - 192524097 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

V G ABBIRAMY

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation ; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Wellorganized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand